Gateway Load Balancer (GWLB)

Explore how Gateway Load Balancer (GWLB) facilitates deep packet inspection and centralized security by routing traffic through virtual appliances using GENEVE encapsulation. Understand its architecture, real-world use cases for compliance and security, and how to implement GWLB with AWS CLI and CloudFormation to secure multi-VPC environments.

If we've ever needed to inspect, monitor, or filter traffic flowing between resources in the AWS environment, we know how complex it can get, especially when scaling across multiple VPCs. **Gateway Load Balancer (GWLB)** is a purpose-built solution that makes deep packet inspection possible, streamlined, and scalable.

In this lesson, we'll explore why GWLB is a powerful tool for network security and compliance, how it works under the hood using encapsulation and endpoints, and when to choose it over other inspection strategies like traffic mirroring.

Why does Gateway Load Balancer matter?

Before GWLB, developers and network architects often relied on Application Load Balancer (ALB) and Network Load Balancer (NLB) for routing and load distribution. While these services are robust for their intended use cases, they fall short regarding advanced traffic inspection.

- ALB operates at layer 7 and is optimized for HTTP/S-based traffic for web applications that need content-based routing. However, it terminates the request and cannot transparently forward traffic for deep inspection.

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- NLB, on the other hand, works at layer 4 and offers ultra-low latency performance. It supports TCP and UDP traffic but lacks mechanisms for embedding metadata or enabling transparent, inline inspection. There's also no built-in support for encapsulating traffic for routing through third-party appliances.

GWLB solves these limitations by operating at layer 3 and using GENEVE encapsulation to forward traffic to and from third-party virtual appliances transparently. It enables full control over traffic flow for inspection without altering packet headers or requiring changes to application code, uniquely suiting it for security use cases, deep packet inspection, and compliance monitoring.

> **Note:** GWLBs are particularly useful in environments with multiple VPCs or accounts, where centralized security and inspection policies are required. It enables consistent control while maintaining a loosely coupled architecture.

How GWLB works: Architecture and data flow

The GWLB is the entry point for inspecting traffic. It forwards packets to a target group of appliance instances using GENEVE encapsulation, which adds metadata to the packet.

- **Target group of virtual appliances:** These are EC2 instances running virtual firewalls, IDS, or other security tools that function like physical network appliances but are software-based and run in the cloud. These "virtual appliances" mimic the behavior of traditional hardware firewalls, intrusion prevention systems, or content filtering systems but can scale dynamically and be centrally managed within AWS. Each appliance must support GENEVE encapsulation to decode the original traffic, inspect it, and t return it.

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- **GWLB endpoint (GWLBe):** This is a PrivateLink-based endpoint in the VPC. We update our route table to forward traffic to this endpoint. From here, traffic enters the GWLB, is processed by appliances, and is returned to its destination. To make this concrete, imagine a web application hosted in a private subnet of a VPC that needs to access external services securely. Instead of routing this traffic directly through a NAT gateway or internet gateway, we can redirect it through a GWLB endpoint. This redirection changes the VPC's routing table route to point to a GWLB endpoint.

  Once traffic reaches the endpoint, it flows into the GWLB, which is encapsulated using GENEVE and forwarded to a fleet of EC2-based virtual appliances (such as a firewall or intrusion detection system). The appliance inspects the traffic, checking it for malware, policy violations, or sensitive data, and sends it back to the GWLB. After decapsulation, the traffic is routed to its intended destination, maintaining flow fidelity and security.

  This design allows developers and security teams to insert sophisticated inspection tools into the network path without redesigning application logic or network infrastructure.

Example: Inspecting traffic with GWLB AWS CLI

We'll start by creating a GWLB endpoint in a subnet using the AWS CLI. We'll define the VPC, endpoint type, service name, and the subnet where this endpoint will reside. This allows the redirection of VPC traffic for inspection.

main.sh C++

```
aws ec2 create-vpc-endpoint \
  --vpc-id vpc-0abc1234de567890f \
  --vpc-endpoint-type GatewayLoadBalancer \
  --service-name com.amazonaws.vpce.us-east-1.vpce-svc-0123456789abcd
  --subnet-id subnet-0123abcd4567efgh8
```

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Creating a GWLB endpoint using AWS CLI

After creating the endpoint, we need to update the route table to send specific traffic—like all outbound traffic (0.0.0.0/0)—to the GWLB endpoint. This ensures that all such traffic is inspected before leaving the VPC.

main.sh Shell-20

```
aws ec2 create-route \
  --route-table-id rtb-0123456789abcdef0 \
  --destination-cidr-block 0.0.0.0/0 \
  --vpc-endpoint-id vpce-0123456789abcdef0
```

Route table entry to send traffic to the GWLB endpoint

Example: Defining a GWLBe in CloudFormation

For infrastructure as code (IaC) implementations, we can define a GWLBe i CloudFormation template. This provides a repeatable and consistent way t deploy inspection capabilities across environments.

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main.sh C++

```
Resources:
  GwlbEndpoint:
    Type: AWS::EC2::VPCEndpoint
    Properties:
      ServiceName: !Sub "com.amazonaws.vpce.${AWS::Region}.vpce-svc-0
      VpcId: vpc-0abc1234de567890f
      VpcEndpointType: GatewayLoadBalancer
      SubnetIds:
        - subnet-0123abcd4567efgh8
```

Defining a GWLBe in CloudFormation

These configurations allow developers to programmatically integrate GWLB into their infrastructure and direct traffic securely for inspection.

GWLB: Real-world scenarios

The Gateway Load Balancer (GWLB) excels at integrating traffic inspection tools into our network. We can insert security or monitoring appliances without changing our applications or existing network configurations. This is particularly useful for environments that need to centralize security policy enforcement or gain deep network visibility across multiple VPCs and AWS accounts.

- **Security enforcement use cases:** GWLB supports centralized firewalling by allowing traffic from multiple VPCs to flow through a shared fleet of firewalls, significantly simplifying policy management and reducing replication. In multi-account environments using AWS Organizations, spoke accounts can route traffic to a centralized security VPC for inspection, maintaining separation of concerns while enhancing oversight. Additionally, inline threat

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filtering allows appliances behind GWLB to block or redirect malicious or unauthorized traffic based on real-time analysis.

- **Compliance and observability:** Regulated industries often rely on deep packet inspection (DPI) to meet standards like PCI-DSS and HIPAA, which GWLB facilitates by routing traffic through IDS or IPS systems. For data governance, developers can route outbound traffic through DLP appliances to prevent leaks of PII or intellectual property without changing application code. GWLB also aids observability by consolidating traffic logging, enabling clearer visibility and faster incident response across distributed workloads.
- **Functional flexibility:** Compared to traffic mirroring, which is passive, GWLB enables active inline inspection where appliances can influence traffic flow directly. This flexibility makes it suitable for security enforcement rather than just monitoring. Additionally, because appliances can be added or removed dynamically from target groups, GWLB supports modular scalability independent of the application tier. Developers can also build advanced service chaining by routing traffic through multiple inspection layers using route tables and GWLB endpoints.
  By combining GWLB with tools like CloudWatch, Firewall Manager, and VPC Flow Logs developers and security engineers can build a powerful, observability-rich inspection layer for their AWS workloads.

Conclusion

In this lesson, we explored the capabilities and architectural principles of Gateway Load Balancer (GWLB). We learned how GWLB addresses limitations found in traditional load balancers like ALB and NLB by enabling transparent, scalable, and centralized traffic inspection through GENEVE encapsulation. We walked through how GWLB works—from the role of GWLB endpoints to the flow of traffic through virtual appliances—and examined real-world use cases ranging from centr firewalling to compliance-driven inspection strategies.

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Understanding GWLB is important for the AWS Certified Developer – Associate (DVA-C02) exam because it ties directly into topics like secure traffic flow, VPC integration, service interconnectivity, and the use of AWS-native constructs for scaling and securing applications. Mastery of GWLB can help us design solutions that meet security and operational requirements without compromising agility or

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